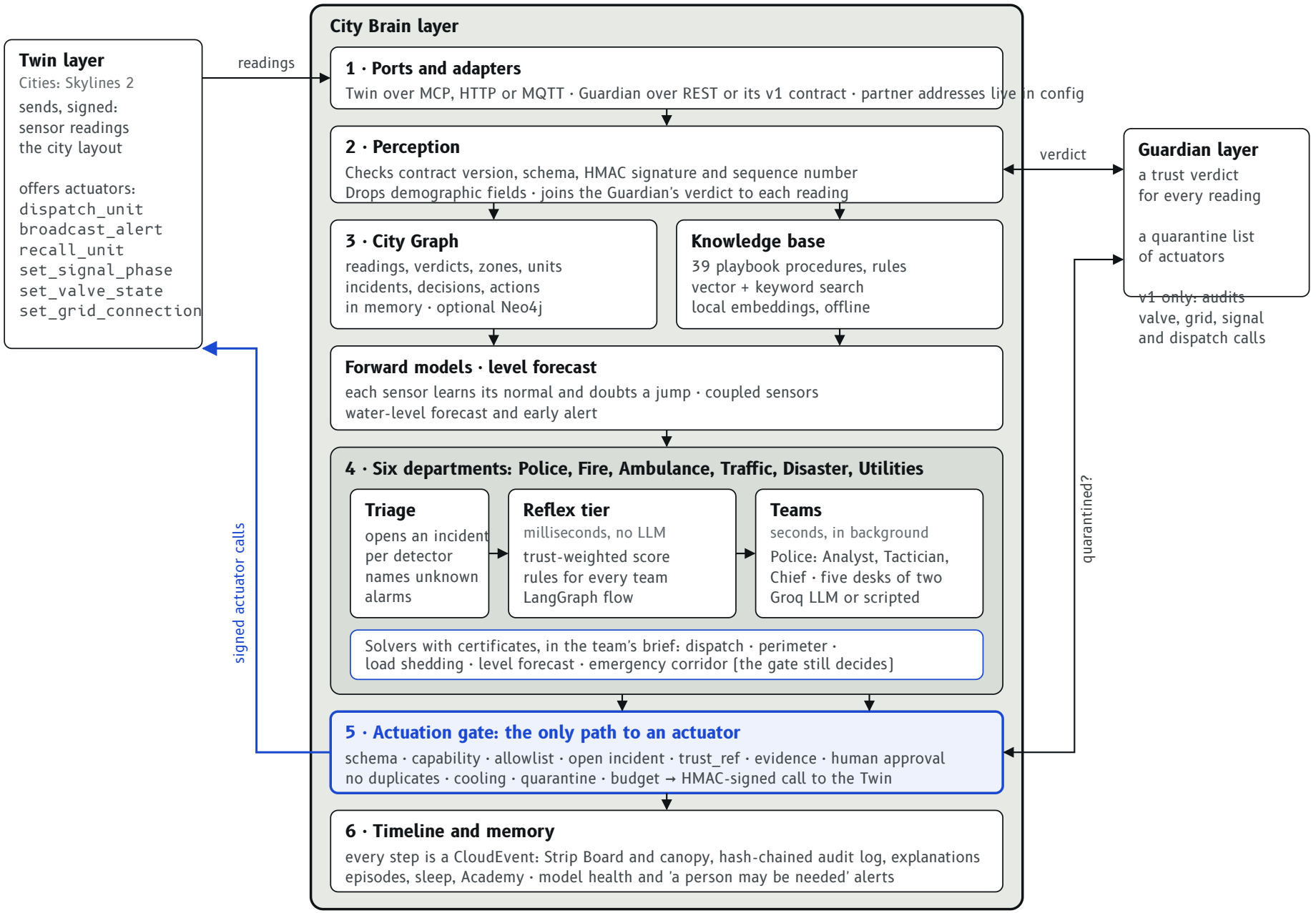
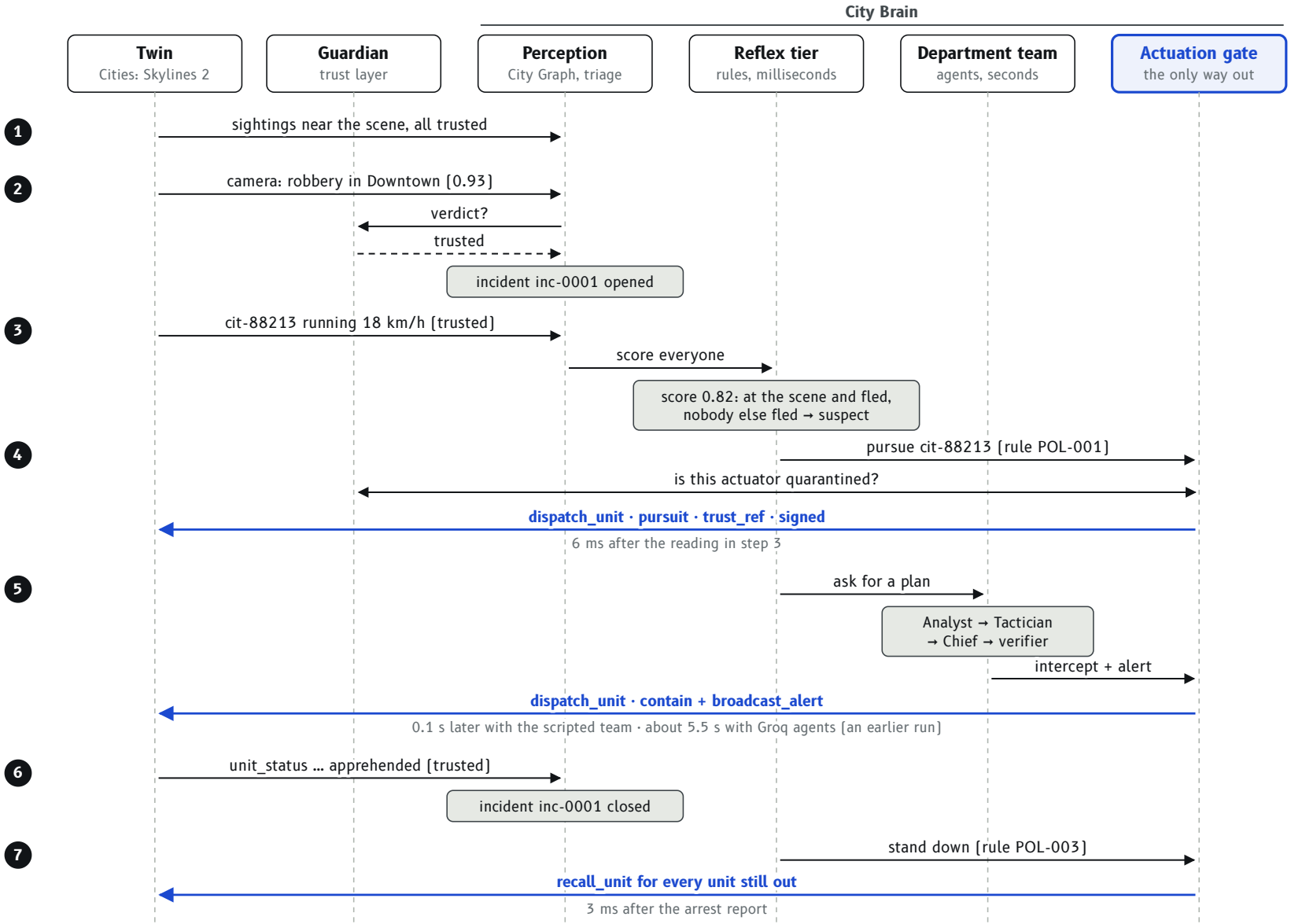
City Brain architecture: data flows down, and actions leave only through the gate

Everything drawn here is built and runs today. Six departments, five solvers. TSYP14 City Brain layer, updated 6 November 2026.



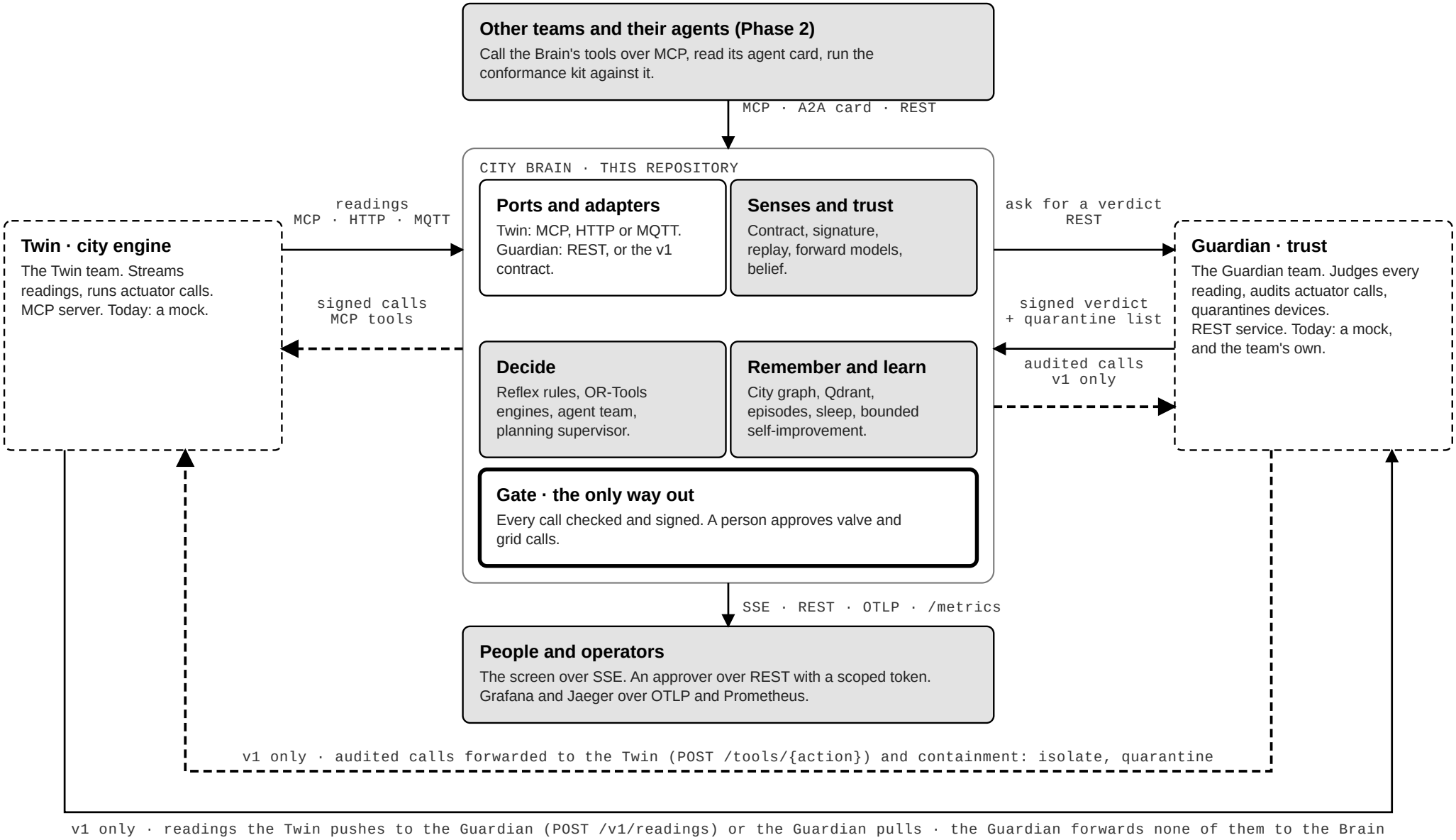
The pursuit leaves 6 ms after the deciding reading; the agents add the rest 0.1 s later

Scenario t14, a crime and a pursuit, run in process on 7 November 2026 against the stand-in Twin and Guardian. Times are from its log.



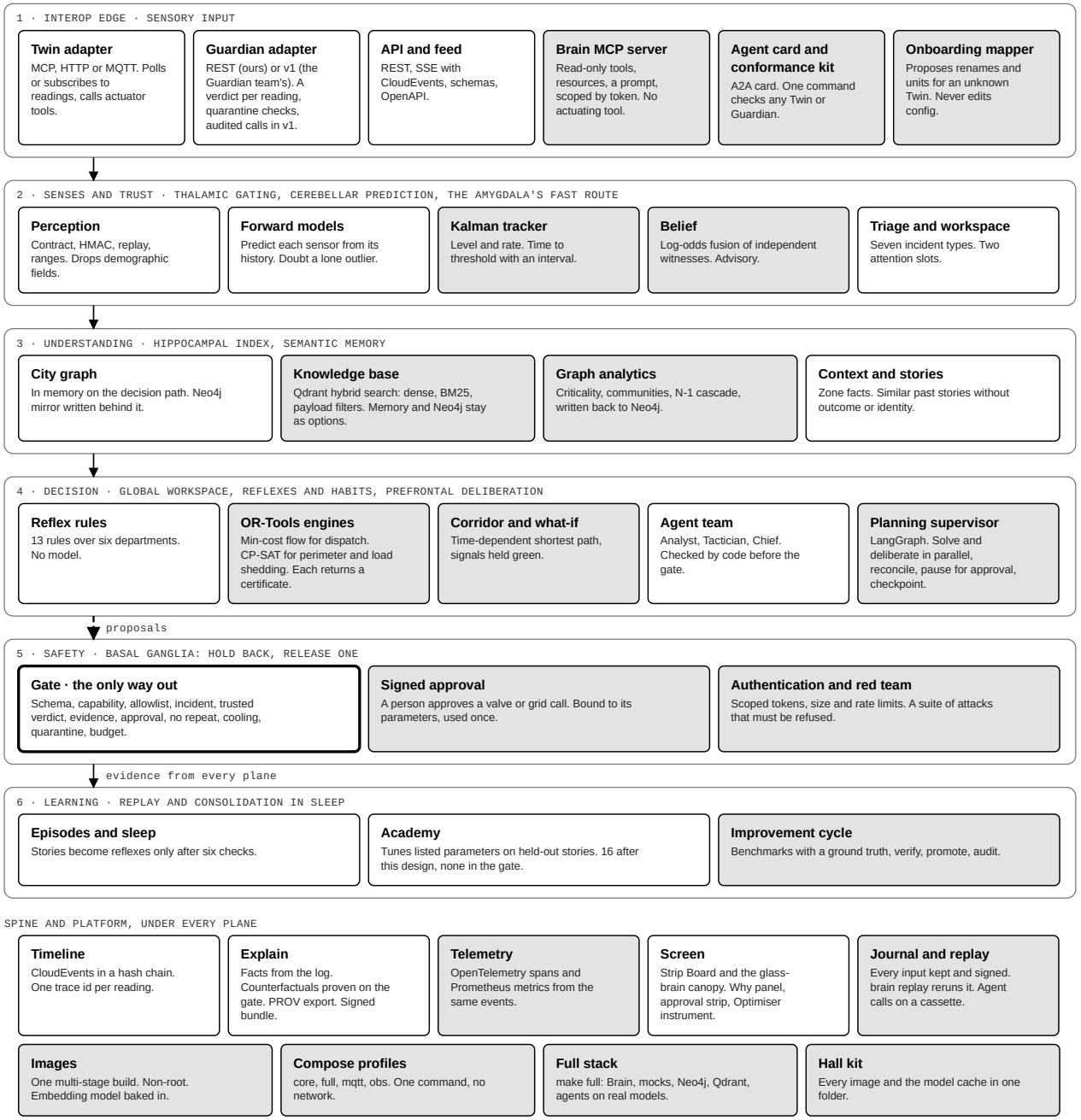
Every arrow is also a timeline event: it appears on the Strip Board screen and in the hash-chained audit log.
If the Guardian answers untrusted in step 2 or 3, nothing is dispatched and the reading is held at the gate with its reason [shown for a flood in scenario t13].
Run against the stand-in Guardian, which the Brain sends each reading to. With the Guardian team's Guardian the Twin also feeds it, and it audits valve, grid, signal and dispatch calls [Figure 9].

The City Brain between its partners



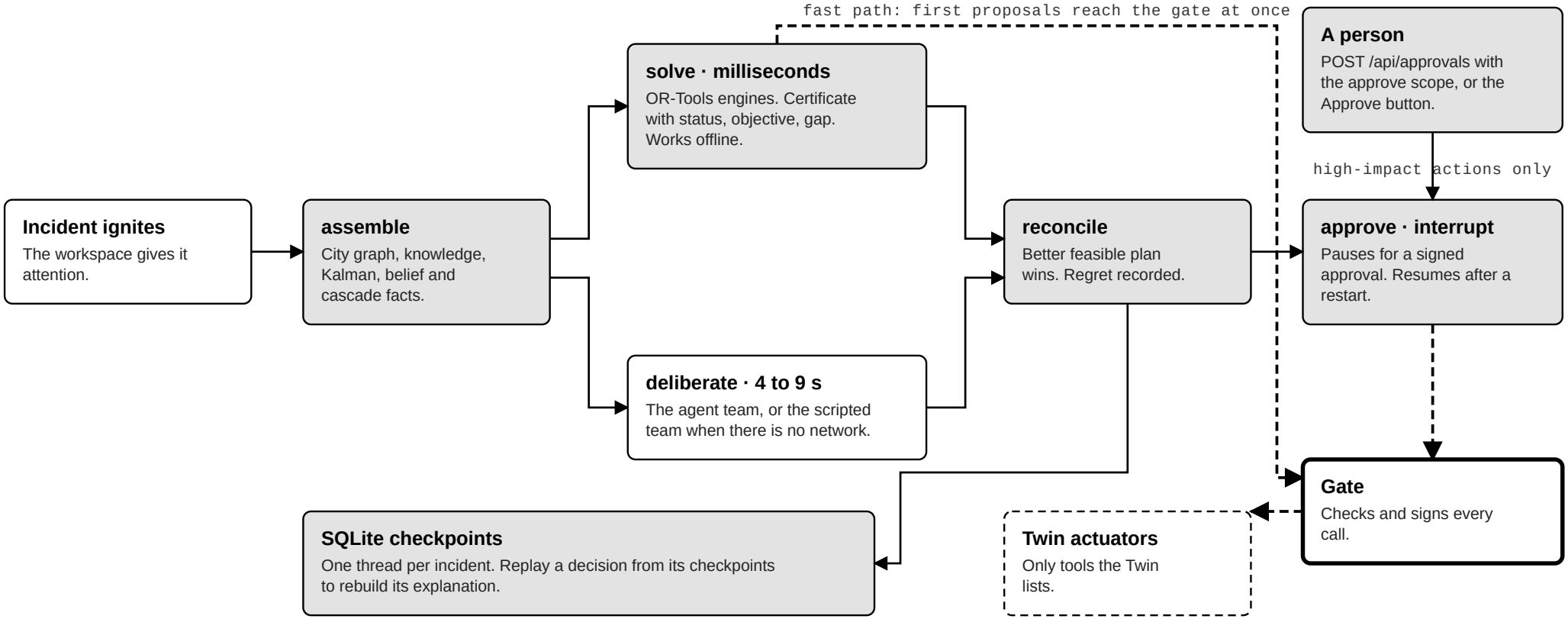
The Brain between the Twin and the Guardian, with the protocol on every link. The two lines under the Brain exist only with the Guardian team's Guardian (Figure 9).

Figure 1. The layered architecture



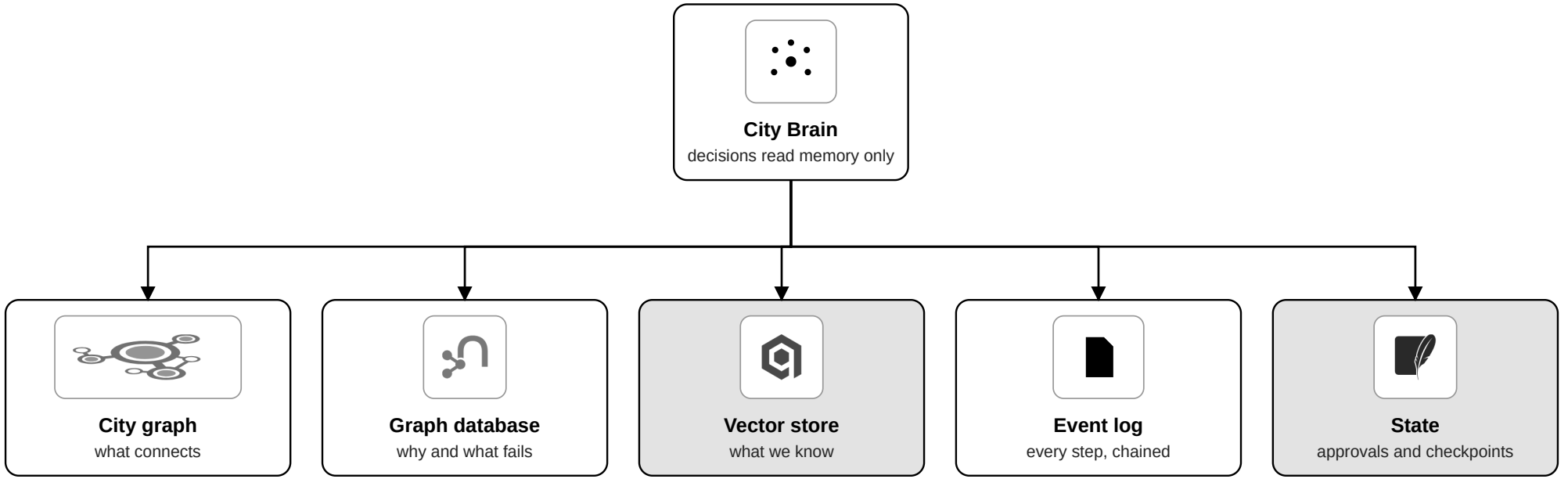
Six planes from the interoperability edge to learning, the shared spine and the platform. A grey box is new in the Convergence design.

Figure 2. The planning graph



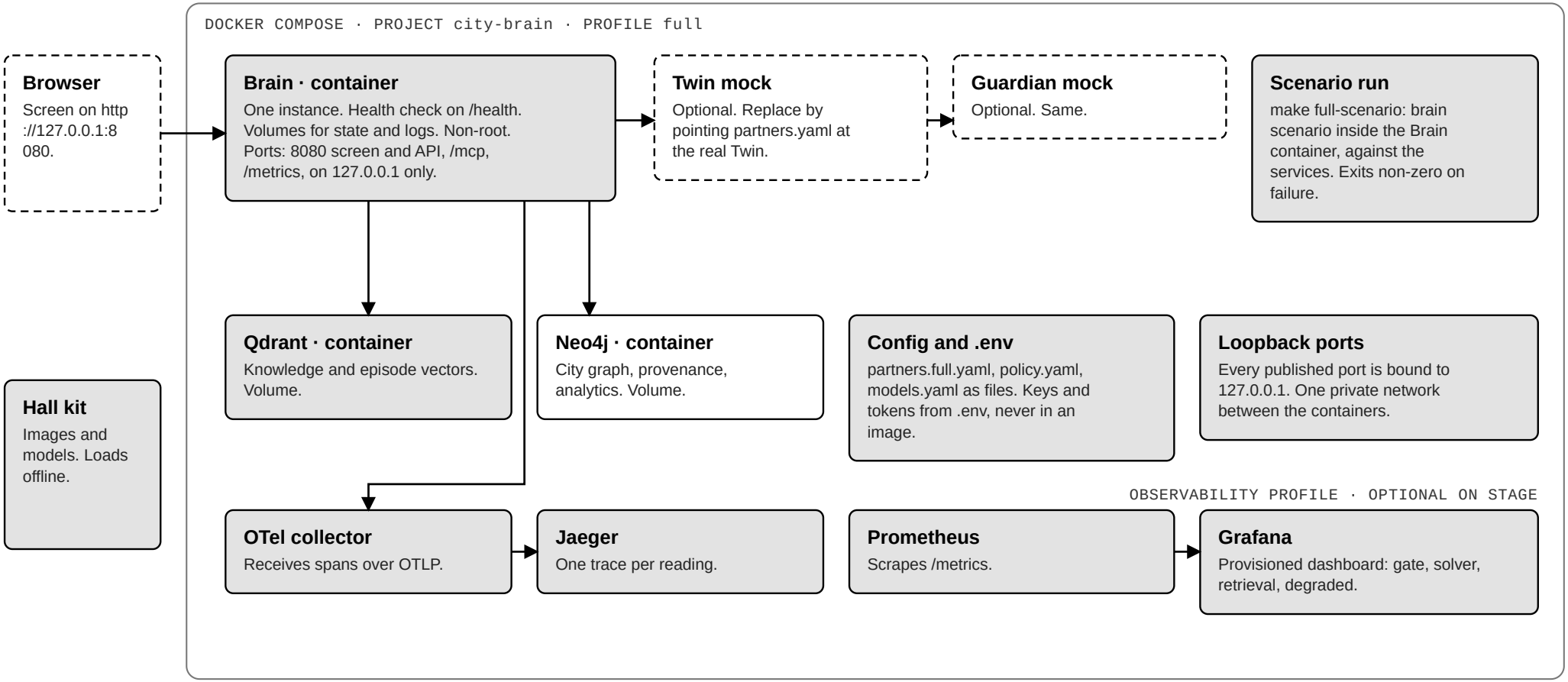
Solve and deliberate in parallel, reconcile, pause for a person, then the gate. Checkpoints sit under the graph.

Figure 3. The memory plane



Five stores, each answering a different question, none on the decision path's critical wait.

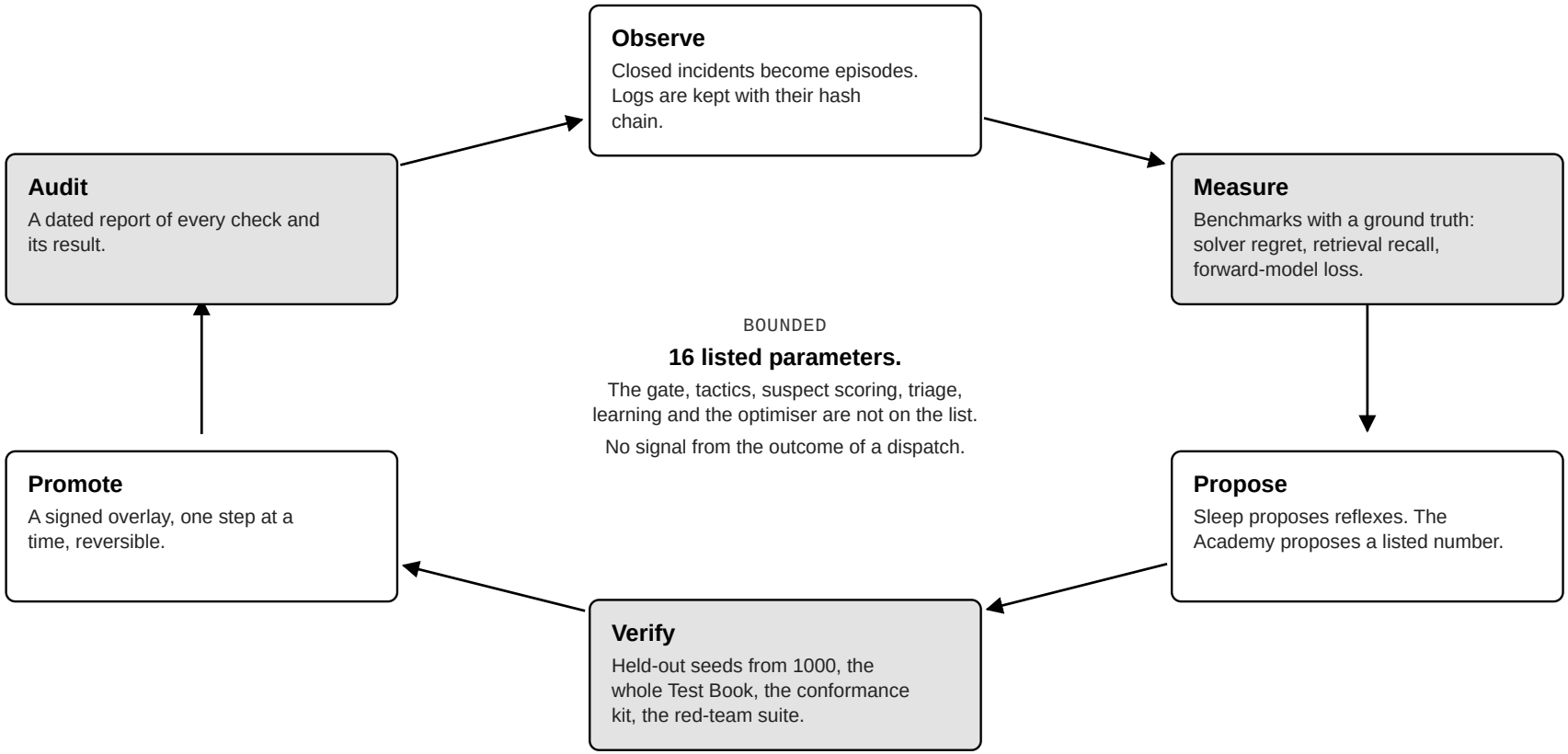
Figure 4. Deployment with Docker Compose



Compose profiles core, full, mqtt, obs run the same images on one laptop. The core profile is the default on stage.

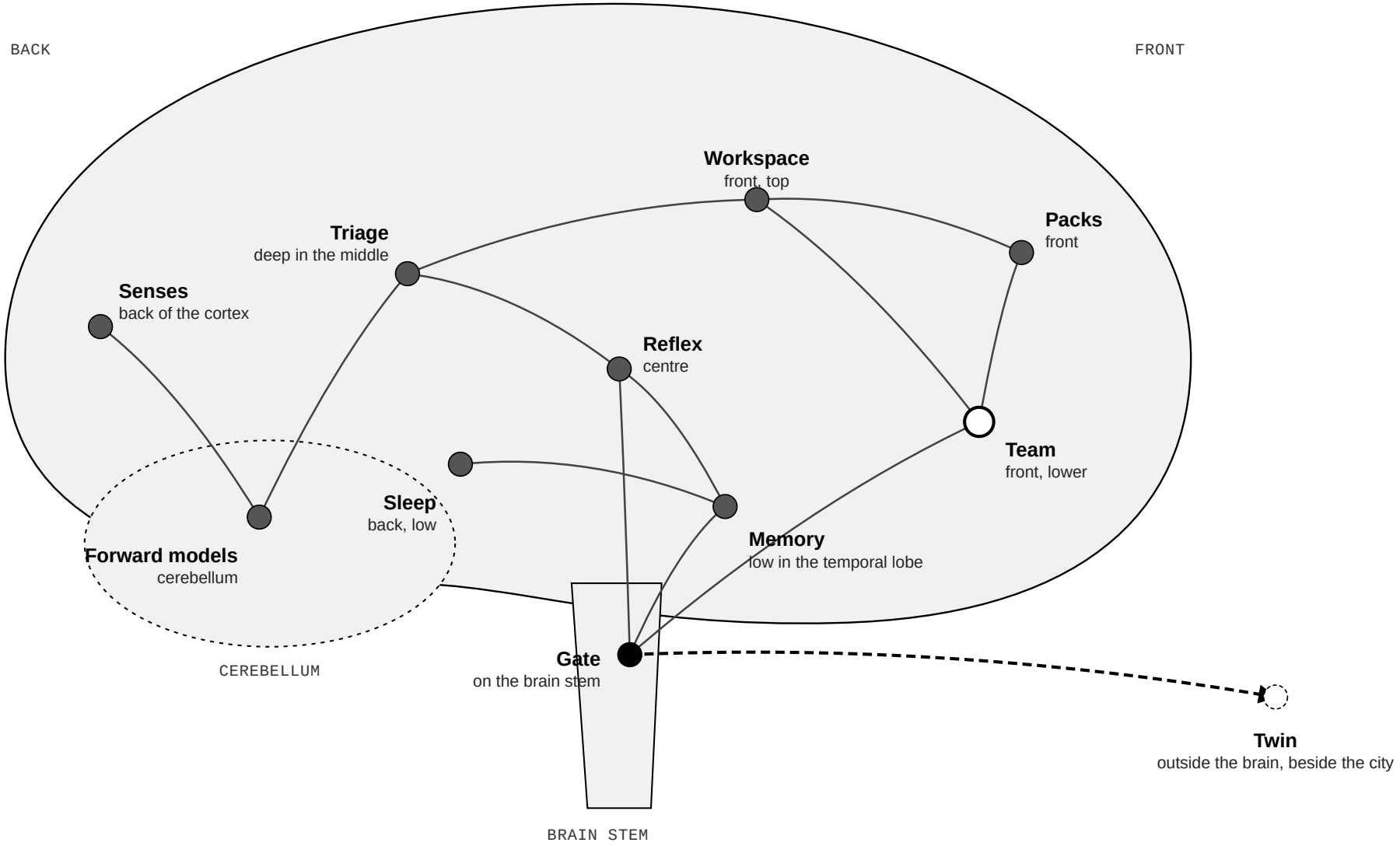
The Brain, its stand-in partners, stores and observability in one Compose project.

Figure 6. The improvement loop



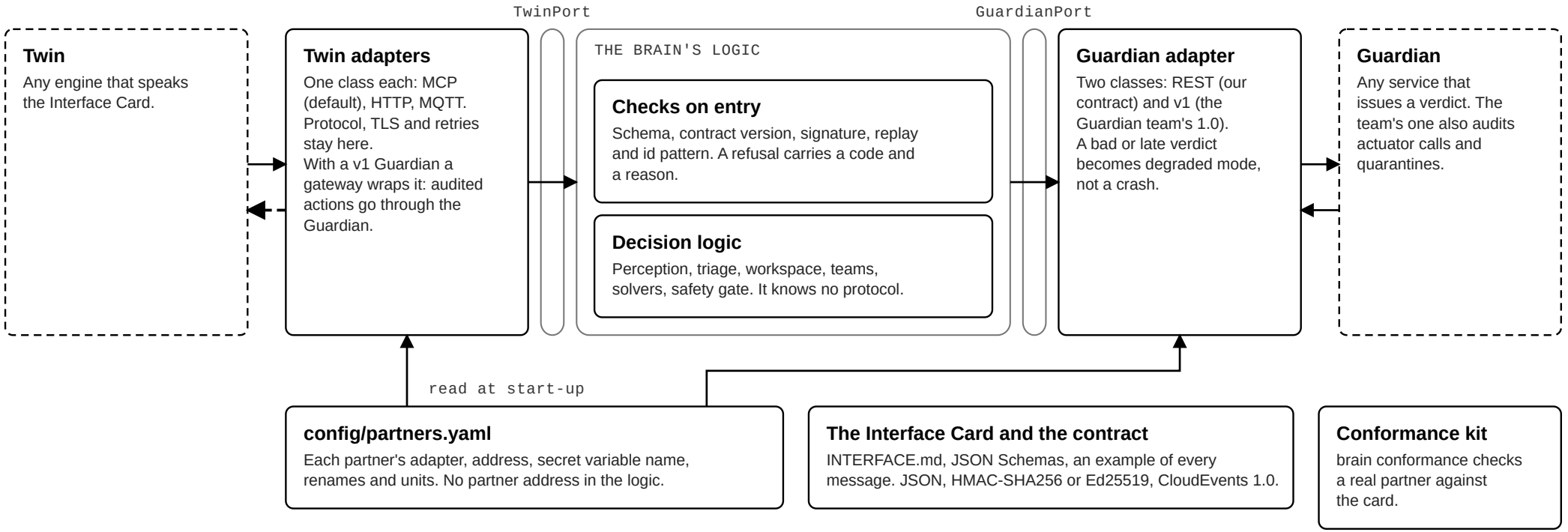
Observe, measure, propose, verify, promote, audit: bounded, reversible, never on the gate.

Figure 7. The glass brain from the side



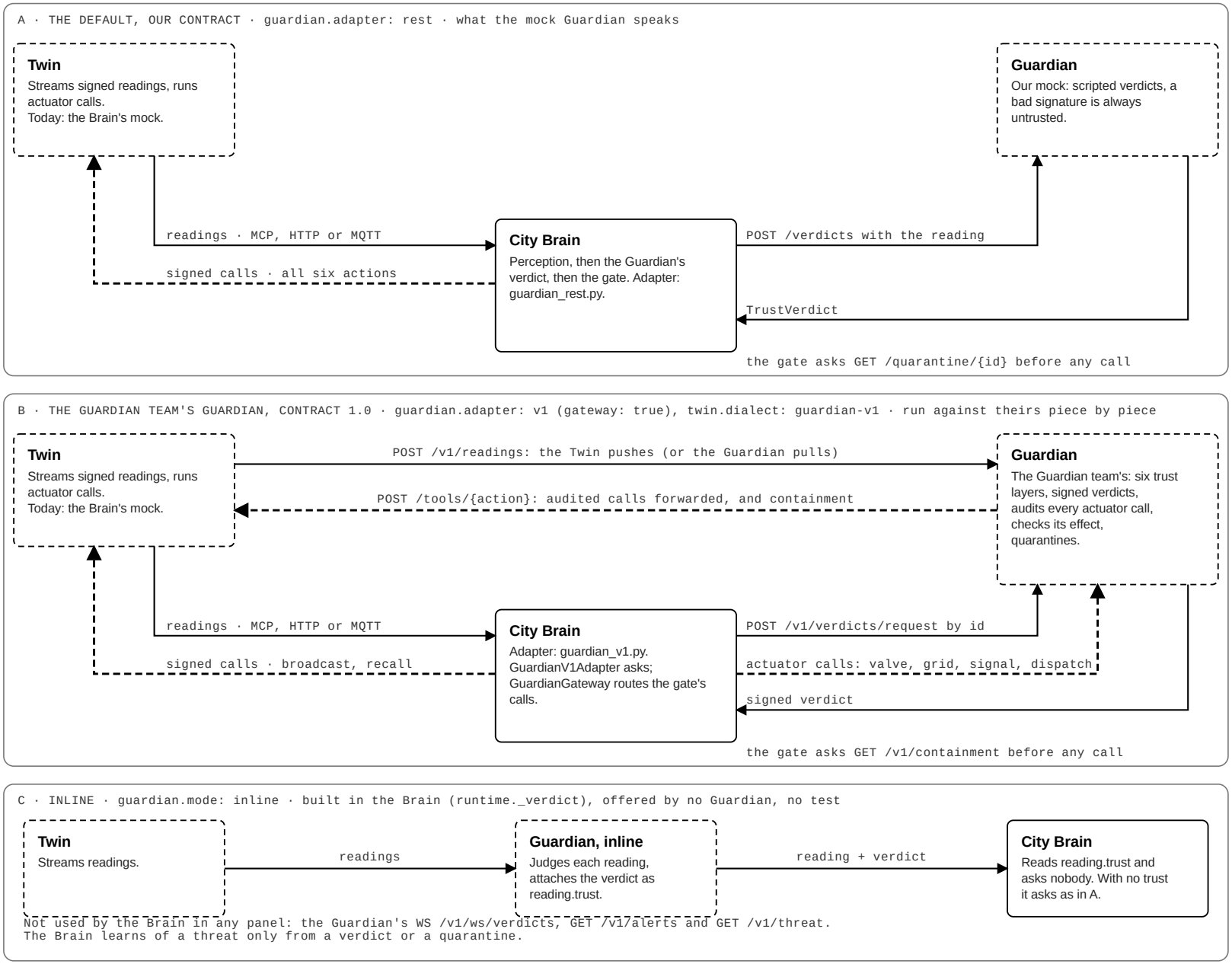
Each region where the screen draws it. The Brain is brain-inspired: it models functions, not a brain.

Figure 8. Where logic ends and partners begin



Two ports, the adapters outside them, and the configuration file, Interface Card and conformance kit that make a new partner a small change.

Figure 9. The Guardian wired three ways



A: our contract, the Brain sends each reading to the Guardian. B: the Guardian team's contract 1.0, the Twin also feeds the Guardian and the Guardian audits actuator calls. C: inline, built in the Brain, offered by no Guardian.